TEXORA · LECTURE SHEET

Reactive and Disperse Dyes

Covalent fixation racing hydrolysis on cotton; solid-solution diffusion into polyester at 130 °C, by carrier, or by thermosol — the two systems that colour most of the world's cloth.

TE-3105 · TEXTILE CHEMISTRY & COLORATION-II · TOPIC 3 · LEVEL 3 TERM 1

1. Reactive dyes — structure of the molecule

A reactive dye is written **S–Col–B–R**: solubilising groups **S** (sulphonates), the chromophore **Col**, a bridging group **B**, and the **reactive group R** that forms a covalent bond with fibre hydroxyls (cellulose) or amino groups (wool). Commercial families are named by R: monochlorotriazine (MCT, e.g. Procion H), dichlorotriazine (DCT, Procion MX), vinylsulphone (VS, Remazol — applied as the masked sulphatoethylsulphone and activated by alkali), difluorochloropyrimidine, and bifunctional dyes carrying two reactive groups (e.g. MCT + VS) for higher fixation.

2. The fixation reaction — and its rival, hydrolysis

Under alkali, cellulose ionises to the nucleophilic cellulosate anion, which attacks the reactive group (nucleophilic substitution on triazine; Michael addition to the vinylsulphone double bond):

$$\text{Dye–Cl (triazine)} + \text{Cell–O}^- \rightarrow \text{Dye–O–Cell} + \text{Cl}^- \quad | \quad \text{Dye–SO}_2\text{–CH=CH}_2 + \text{Cell–O}^- \rightarrow \text{Dye–SO}_2\text{–CH}_2\text{–CH}_2\text{–O–Cell}$$

The same nucleophile exists in water as OH^-, and every fixation bath runs the competing reaction:

$$\text{Dye–R} + \text{OH}^- \rightarrow \text{Dye–OH (hydrolysed dye — can never fix)}$$

Fixation efficiency is therefore never 100%: typical values are 50–70% for monofunctional and 75–90% for good bifunctional dyes. Hydrolysed dye has the same colour but no bond; it must be completely soaped off, otherwise wash fastness and staining fail even though the shade looks right. Fixation rises with alkali dose, temperature (per dye class: DCT cold 30–40 °C, VS warm 40–60 °C, MCT hot 80 °C) and low liquor ratio, while hydrolysis accelerates with pH and standing time — the process window is a race between two reactions of the same dye.

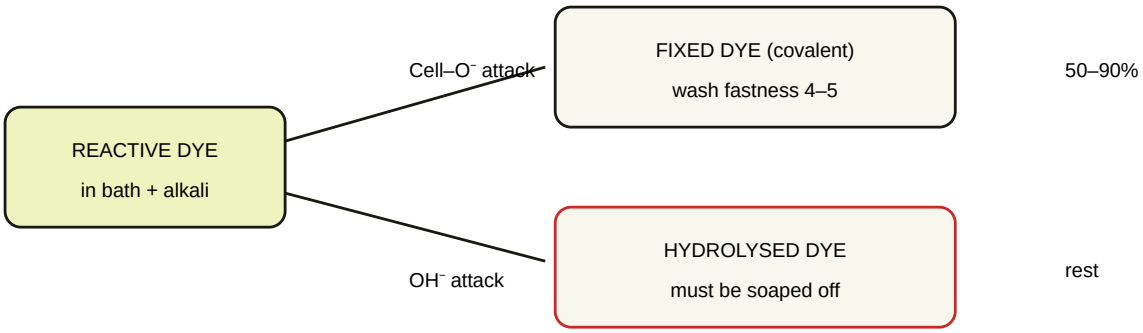


Fig. 1 — The fixation–hydrolysis competition that governs every reactive dyeing.

3. The reactive dyeing cycle (exhaust, cotton)

1. **Exhaustion phase:** dye + electrolyte (NaCl or Na_2SO_4, 30–100 g/L by depth) at neutral pH. Salt screens the negative surface charge of cellulose (zeta potential) so anionic dye can approach; exhaustion reaches 60–80%.
2. **Fixation phase:** alkali dosed progressively (soda ash to pH 10.5–11.5; caustic for some systems). Progressive dosing avoids a fixation strike that would dye unlevelly.
3. **Wash-off:** cold rinse, acid neutralise, hot soaping at the boil with detergent (twice for deep shades), warm rinse — removal of hydrolysed dye is a quality step, not an afterthought.

4. Disperse dyes — the polyester system

Disperse dyes are small, non-ionic, sparingly soluble molecules (azo and anthraquinone) milled with dispersing agent to a stable aqueous dispersion. Polyester is hydrophobic, highly crystalline and has a glass transition near 70–80 °C (wet); below it the polymer chains barely move and dye diffusion is negligible. Three industrial routes open the fibre:

ROUTE	CONDITIONS	NOTES
High-temperature high-pressure (HTHP)	125–135 °C, pressurised machine, pH 4.5–5.5, 30–60 min	Standard for piece and package; highest levelness and fastness
Carrier dyeing	95–100 °C with 3–8 g/L carrier (o-phenylphenol, chlorinated aromatics historically)	Carrier swells polyester and plasticises it, lowering effective Tg; odour/effluent problems — now restricted, used for deep shades and wool blends that cannot take 130 °C
Thermosol	Pad–dry–bake 190–220 °C, 30–90 s (continuous, for	Dye sublimes/diffuses into softened polyester; the cotton is then reactive-dyed

polyester/cotton)

Dyes are graded by **energy (sublimation) class** — low-energy (easy dyeing, poor sublimation fastness), medium, and high-energy (deep shades, best heat fastness). After HTHP dyeing, surface dye and oligomer are removed by **reduction clearing** (sodium hydrosulphite + caustic at 70–80 °C), which is essential for wash and rub fastness in medium–deep shades. Mechanism: dye dissolves in the bath to a minute molecular concentration, partitions into the fibre as a **solid solution**, and diffuses — exhaustion is driven by that partition, not by ionic attraction, so no salt is used and pH is kept mildly acidic to protect dye and fibre.

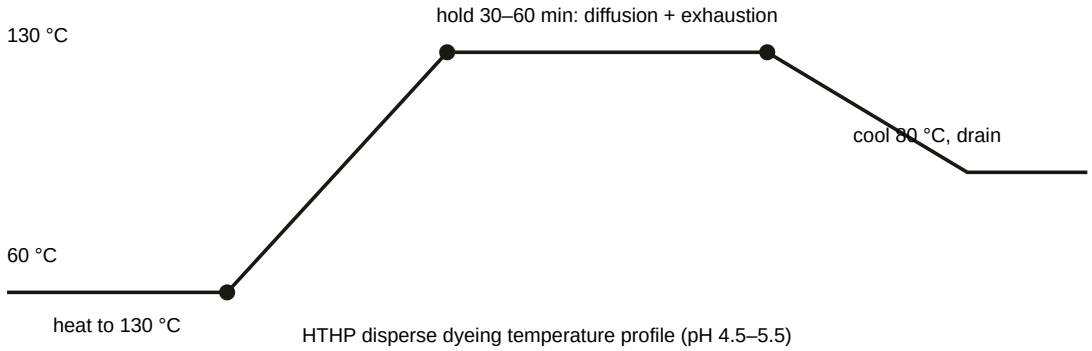


Fig. 2 — Temperature profile of HTHP disperse dyeing of polyester.

5. Contrasts that matter in the mill

FEATURE	REACTIVE ON COTTON	DISPERSE ON POLYESTER
Bond / retention	Covalent bond	Solid solution, van der Waals
Electrolyte	Essential (30–100 g/L)	None
pH	Alkaline for fixation (10.5–11.5)	Acidic (4.5–5.5)
Main loss route	Hydrolysis	Surface dye if not reduction-cleared; sublimation in heat setting
Wash fastness	Excellent after full soaping	Excellent after reduction clearing

6. Glossary — key terms

- **Hydrolysis:** reaction of the reactive group with water/OH^-, destroying fixability.
- **Fixation efficiency:** percentage of applied dye covalently fixed at the end of dyeing.

- **Carrier:** swelling/plasticising auxiliary enabling disperse dyeing at the boil.
- **Reduction clearing:** hydrosulphite/caustic wash removing surface disperse dye.
- **Energy class:** grouping of disperse dyes by molecular size and sublimation fastness.

7. Key points

- Reactive dyeing is a managed race: fixation on Cell–O^- versus hydrolysis by OH^-.
- Salt exhausts, alkali fixes, soaping proves the result.
- Disperse dyeing needs the polyester above its glass transition: 130 °C, or a carrier, or thermosol baking.

8. Summary

Reactive and disperse dyes are the two pillars of modern coloration — cotton by covalent chemistry with an unavoidable hydrolysis loss, polyester by thermally activated solid solution with a clearing step to finish. Their opposite chemistries (alkali versus acid, salt versus none, bond versus solution) are why polyester/cotton blends are dyed in two distinct stages.

9. Practice

1. A reactive dyeing shows exhaustion 75% and fixation efficiency 62%. For 3.0% owf applied dye, calculate (a) dye on fibre before fixation, (b) dye covalently fixed, both as % owf, and (c) the hydrolysed/unfixed dye that wash-off must remove.
2. Why is alkali dosed progressively rather than all at the start of fixation?
3. Explain carrier action in terms of polyester's glass transition, and give two reasons carriers are being phased out.
4. A polyester fabric shows poor wash fastness after HTHP dyeing although exhaustion was 95%. What was most likely skipped?

Answer key

1. (a) 3.0 × 0.75 = 2.25% owf on fibre. (b) 2.25 × 0.62 = 1.395 ≈ 1.40% owf fixed. (c) Unfixed = 3.0 − 1.40 = 1.60% owf (0.75% never exhausted + 0.86% exhausted but hydrolysed) must leave in rinses and soaping.
2. Instant full alkali fixes dye where it sits, including any unlevel distribution from exhaustion, and maximises local hydrolysis. Progressive dosing lets migration continue while fixation builds gradually and uniformly.

3. Carriers are absorbed by polyester, swelling it and plasticising the amorphous regions so chain segments move at 100 °C as they would near 130 °C without carrier; dye diffusion follows chain mobility. Phase-out: toxicity/effluent load (chlorinated carriers), odour retained in fabric, and modern HTHP machines make them unnecessary.
4. Reduction clearing. Exhausted dye includes a surface deposit that is not dissolved in the fibre; without hydrosulphite/caustic clearing it desorbs in every wash.

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